



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A++' Grade Awarded Category - I by UGC

Semester-2

Seat No.

Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25,2022-26

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Date: 26/12/2023

Day: Tuesday

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1 a) Design a flowchart to find greatest of three numbers. 3 CO1, BL3
- b) Write a short note on IDE and their common features. 3 CO1, BL2
- Q.2 a) What is typecasting? Explain its different types with examples. 4 CO2, BL2
- b) On a birthday party, there are some family members and friends to be invited. Family members are sure to be invited and their name cannot be changed whereas, there are some friends whose name could be updated. Write a python program containing a single record of all the invitees having 5 family members and list of 5 friends. Later modify one of the friends name and display the members. 4 CO2, BL3

- Q.3 a) Write a python program that accepts vehicle registration number from the user and prints whether the given input is valid or not. The vehicle registration number must be in 'AABBCCDDDD' format where,
AA - state code(alphabets)
BB- RTO city code(numbers)
CC- series (alphabets)
DDDD- numbers. (Hint : Use string inbuilt methods). 4 CO3, BL3
- b) Write a program to differentiate between map and filter functions in python. 4 CO3, BL3
- Q.4 a) Illustrate the use of constructors and destructors in class. 4 CO4, BL4
- b) Write a program to create a base class Flower with attributes such as name, color, and behavior show. Create a child class Rose with the attribute of petalcount and method to print the Flower's name, color and petalcount. 4 CO4, BL3

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2021-25

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Batch: 2021-25 (Sem II)

Backlog October/Odd 2022 Examination

Date: 09/01/2023

Day: Monday

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER**Instructions:**

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1** a) Illustrate different phases of Computational thinking with Example. 3 CO1, BL4
- b) Write an algorithm to find factorial of a number using recursion. 3 CO1, BL3
- Q.2** a) Predict the output and justify your answer: 3 CO2, BL3
- i) `-67%7`
 - ii) `6.6//6`
 - iii) `(90-70) * 10/5`
 - iv) `not False`
 - v) `type((4,5))`
 - vi) `5*1**2`
- b) Describe any two list operations and list methods. Write a Python program to check whether an element is present in the list or not using membership operator. 3 CO2, BL2
- Q.3** a) Write a Python program to prompt a number from user and find reverse of number using recursion 4 CO3, BL3
- b) Discuss about any two methods for passing parameters to a function with examples. 4 CO3, BL2

- Q.4** a) Create a class named School having class attribute as SchoolCity. Instance attributes as StudName and StudStd. The Student class should have a method as SetAge(), that prompts age from user and displays the age. 5 CO4, BL5
- b) Create a child class that will inherit all the features of the School class used in Q.4 (a) and have additional attribute as SchoolBranch. Also, create show() method to display all the features of School class. 5 CO4, BL5

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Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26, 2021-25

Semester: II

Course: Programming and Problem Solving

Course Code: 0705210206

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination

Date: 17/07/2023

Maximum Marks: 30

Day: Monday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
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3. Figures to the right indicate full marks.

Q.1 a) Design a flowchart and an algorithm to find greatest of two numbers. 3 CO1, BL3

b) Illustrate concept of computational thinking with real life example. 3 CO1, BL2

Q.2 a) The class has student roll numbers from 1 to 20. 4 CO2, BL3

$G1 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$G2 = \{5, 6, 7, 8, 9, 10, 11, 12, 13\}$

$G3 = \{1, 2, 6, 7, 11, 12, 15, 16, 20\}$

G1 is a set of roll numbers who play cricket.

G2 is a set of roll numbers who play football.

G3 is a set of roll numbers who play chess.

Write a python program to answer the following questions related to above mentioned sets:

(i) Display the roll number of students who play outdoor games only.

(ii) Display the roll number of students who play all the games.

b) In a multi national company, the HR wants to keep the records of recruited employees. He has to maintain the data as id, and name, age, salary, address. If new employee is recruited he should not get the same emp id. Write a python program to add new employee and retrieve all the data of the existing employees based on his id. (Hint: Use dictionary) 4 CO2, BL3

- Q.3 a) Write a program to find whether a number entered by the user is a spy number or not using user-defined function. (Hint: spy number is a number if the sum and product of its digits are equal e.g 132 is a spy number) 4 CO3, BL3
- b) Write the difference between a user defined function and lambda function with example in python. 4 CO3, BL2
- Q.4 a) Write a python program to create an object of a DOG class with attributes as color, breed and name. Use destructor to release the memory. 4 CO4, BL3
- b) Describe in detail the concept of inheritance and method overriding with examples. 4 CO4, BL3

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination



Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Seat No.

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Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: II

Course: Communication Skills

Course Code: 0705210217

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination

Date: 17/07/2023

Maximum Marks: 30

Day: Monday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- a) Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
- b) Write your seat number in words and figures in the Answer sheet provided to you.
- c) Do not copy or help anyone in person or by using signs/ digital gadgets while you write the exam/ while you start the exam/ while you end the exam.
- d) Figures to the right indicate maximum marks.
- e) Draw the diagram wherever necessary.

Q1. Attempt all the questions and identify the suitable answers.

10*1=10 marks

(CO5 BL1)

1. Which is the suitable format for dates in a cover letter?

- a. 7th May, 2023
- b. May 7, 2023
- c. 7/5/2023
- d. 5/7/2023

(CO4 BL2)

2. How are consonant sounds classified with reference to origin of sound.

- a. Place of Articulation
- b. Manner of articulation
- c. Structure of articulation
- d. Synthesis of articulation

(CO1 BL1)

3. Who Propounded the theory of ACT theory?

- a. John Elder
- b. Sigmund Freud
- c. John Anderson
- d. Paul Elder

(CO4 BL2)

4. Identify different parts of a syllable structure:

- a. allophone, rime and nucleus
- b. Onset, nucleus and coda
- c. Elision, coda and rime
- d. Coda, prosody and rime

(CO3 BL1)

5. Horizontal communication flows through.....
- Face to Face communication
 - Telephonic communication
 - Periodic communication
 - All of the above

(CO1 BL1)

6. The Johari's window model of communication is by.....
- James Anderson and Judith Butler
 - Edward Peters and Claude Shannon
 - Joseph Luft and Harry Ingram
 - Jacques Lacan and Henry James

(CO2 BL1)

7. Identify which Human information processing is not a part of ACT theory.
- Serial processing
 - Parallel processing
 - Linear processing
 - Convergent processing

(CO3 BL1)

8. In presentation which font style is not used:
- Underline
 - Outline
 - Italics
 - Bold

(CO5 BL1)

9. Which of the following is not a resume format?
- Chronological
 - Portfolio
 - Functional
 - Combination

(CO2 BL2)

10. Mention the main quadrants in Johari's Window model of Communication.
- Association, disassociation, transcendence and enlightenment
 - Id, ego, superego and latent
 - Open area, hidden area, blind spot and unknown
 - Open area, altruism, persuasion and unknown

3*5=15 Marks

Q2. Attempt any three questions from the following.

- Describe the function and significance of email writing.
- State the principles to be implemented for an impactful presentation.
- Elaborate on the elements of an effective communication cycle.
- Determine the key features of Indian English.

(CO5 BL2)

(CO3 BL2)

(CO1 BL3)

(CO2,4 BL4)

5*1= 5 Marks

Q3. Attempt any one question from the following.

- Examine Shannon and weaver model of Communication along examples and a neat diagram of the model.
- Compare the major types of resumes and determine the key components that make a successful resume.

(CO1, 2, 3 BL4)

(CO4, 5 BL4)

Batch: 2022-26 (R) and 2021-25 (B) (Sem II)
Regular and Backlog April/Even 2023
Examination



SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act, 1956)
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Institute:

(0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Seat No.

Programme:

(070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch:

2022-26

Semester:

I

Course:

Course Code: 0705210114
Programming and Problem Solving

Batch: 2022-26 (Sem I)

Backlog April/Even 2023 Examination

Date: 26/06/2023

Day: Monday

Maximum Marks: 30

Time: 10:00 - 11:00

Instructions:

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- Q.1 a) Design an algorithm to find whether a number is odd or even. 3 CO1, BL3
b) Write a short note on Computational Thinking. 3 CO1, BL2
- Q.2 a) Create a list of 5 friends in python. Display the names of all friends along with their length of the names separately. 3 CO2, BL3
b) Write a program to create a game such that, when a player enters his date of birth, the program computes single digit sum of it and displays a message based on the sum as follows: if sum is 1, 2 or 3: You are Intelligent, 4, 5 or 6: You are Cool, 7, 8 or 9: You are Gorgeous. 3 CO2, BL3
- Q.3 a) Write a program to find the multiplication table of a number using a recursive function. 4 CO3, BL3
b) Write a short note on modules in python and show the use of any predefined module. 4 CO3, BL2
- Q.4 a) With help of an example, demonstrate the use of "__init__" (constructor) and "__del__" (destructor) in python class. 4 CO4, BL4
b) Write a program to create a base class STUDENT with attributes such as name, age, and behavior display. Create a child class BTech-Student with the attribute of semester and method to print the student's name, age and semester. Explain Inheritance. 6 CO4, BL3

Batch: 2022-26 (Sem I)
Backlog April/Even 2023 Examination

Programme:

SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
(070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Seat No.									
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Batch:

Semester:

Course:

Course Code:

Date: 23/01/2023

Day: Monday

2022-26

I

Programming and Problem Solving

0705210114

Batch: 2022-26 (Sem I)

Regular/October/Odd 2022 Examination

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
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3. Figures to the right indicate full marks.

- Q.1 a) Design an algorithm for finding the sum of digits of a 3-digit number. 3 CO1, BL3
- b) Write a short note on Data Representation in computers. 3 CO1, BL2
- Q.2 a) Create a Dictionary with 5 elements in python with keys as "courses" and values as "number of students enrolled". Display all values and keys separately. 3 CO2, BL3
- b) Write a python program to check if a 3-digit number entered by the user is Armstrong number or not. 3 CO2, BL3
- Q.3 a) Write a program to find the factorial of a number using a recursive function. 4 CO3, BL3
- b) Describe keyword argument and default argument with an example. 4 CO3, BL2
- Q.4 a) Explain the usage of constructor and destructor in python class. 4 CO4, BL1
- b) Write a program to create a base class BIRD with attributes such as color, weight, and behavior Fly. Create another two child classes peacock and eagle with the attribute of flying height and method to print flying height. Explain Method Overriding. 6 CO4, BL3

Institute:

Programme:

Batch:

Semester:

Course:

Course Code:

Date: 21/01/2023

Day: Saturday

(0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
(070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

2022-26

I

Communication Skills

0705210104

(Established under Section 3 of the UGC Act, 1956)
Re-accredited by NAAC with 'A' Grade (3.58/4) Awarded Category - I by UGC

Seat No.

Batch: 2022-26 (Sem I)
Regular October/Odd 2022 Examination

Maximum Marks: 30

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

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- Figures to the right indicate maximum marks.
- Draw the diagram wherever necessary.

Q1. Attempt all the questions and identify the suitable answers.

10*1=10 marks

- If you didn't know the recipient's name, how would you address the person? (CO5 BL1)
 - Dear Mr./Mrs.
 - Dear Sir or Dear Madam.
 - My dear.
 - Hello
- Identify the features of Indian English. (CO4 BL2)
 - The sounds /i/ and /ɪ:/ are not distinguished
 - Indian English pluralizes mass nouns.
 - Indian English uses neutral vowels.
 - Indian English aspirates the /p/ and /f/ sounds.
- Who invented Johari's Window model of Communication? (CO1 BL1)
 - Joseph Luft and Harrington Ingham
 - Sigmund Freud and Jacques Lacan
 - Michel Foucault and Joseph Henry
 - Julia Kristeva and Judith Butler
- Identify different parts of a syllable structure: (CO4 BL2)
 - allophone, rime and nucleus
 - Onset, nucleus and coda
 - Elision, coda and rime
 - Coda, prosody and rime

5. The communication process consists of several steps. In..... the receiver interprets the message and translates into meaningful information. (CO1 BL1)

- a. Feedback
- b. Decoding
- c. Noise
- d. Medium

6. The verbal elements involved in presentation are..... (CO3 BL1)

- a. Knowing the audience
- b. Linguistics
- c. Gestures and body language
- d. Spatial features

7. The method of transmitting a message is known as..... (CO2 BL1)

- a. Gemination
- b. Communication channel
- c. Content
- d. stream

8. What are the types of Interviews? (CO3 BL1)

- a. Face to face
- b. Behavioral
- c. Telephonic
- d. All of these

9. The..... is a part of a research article. (CO5 BL1)

- a. Abstract
- b. Salutation
- c. Greetings
- d. Subject line

10. Which of the factors do not enhance listening skills? (CO2 BL1)

- a. Constant interruptions.
- b. Attention
- c. Clear perception
- d. Summarization of the message

Q2. Attempt any three questions from the following.

- 1. Explain different types of resumes. (CO5 BL2)
- 2. State the responsibilities of a sender in process of Communication. (CO2 BL2)
- 3. Elaborate on barriers to Communication. (CO1 BL3)
- 4. Analyze different types of Communication along with real life implications. (CO3,4 BL4)

5*1= 5 Marks

Q3. Attempt any one question from the following.

- 1. Enumerate the key aspects involved in Communication Cycle along with the diagram with a suitable example of how the cycle can be applied in an interview. (CO1, 2, 5 BL4)
- 2. Describe the set of principles to be followed to create a Presentation along with its implementation. (CO3, 4 BL4)



Seat No.

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Institute: (0705)SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2022-26

Semester: I

Course: Programming and Problem Solving

Course Code: 0705210114

Date: 24/12/2023

Maximum Marks: 30

Day: Sunday

Time: 10:00 - 11:00

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

1. Attempt all questions. Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
2. Make suitable assumptions wherever necessary and figures to the right indicate full marks.

- Q.1 a) Write a Python program to prompt the user to enter a number, multiply and divide the number by 4 without using the arithmetic operators and display the output. 3 CO1, BL3
- b) Describe in detail computational thinking with example. 4 CO1, BL2
- Q.2 a) Write a short note on break and continue statement with example. 3 CO2, BL3
- b) Write a python program to prompt a number from the user and find whether the given number is multiple of 5 and 8 or not using ternary operator. 4 CO2, BL3
- Q.3 a) Write a python program to create a function containing list of five friends. The function should search for the friend's name, if the name is not in the list, add the friend's name in the list and returns the new list or else returns the list as it is. 4 CO3, BL3
- b) Write the advantages of using lambda function over the user-defined functions. 4 CO3, BL3
- Q.4 a) Write a program to create a class Flower with attributes name, color, and behavior show. 4 CO4, BL3
- b) Differentiate between constructors and destructors of python in a class. 4 CO4, BL4

Programme: (070521) BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE AND ENGINEERING)

Batch: 2023-27

Semester: I

Course: Communication Skills

Course Code: 0705210104

Date: 22/12/2023

Day: Friday

Maximum Marks: 30

Time: 13:30 - 14:30

NOTE : DO NOT WRITE ANYTHING ON THE QUESTION PAPER

Instructions:

- Use Ball Pen/ Ink Pen (Blue only) to answer all questions.
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- Draw the diagram wherever necessary.

Q1. Attempt all the questions and identify the suitable answers.

10*1=10 marks

1. Identify the different types of memos
- Request memo and Confirmation memo
 - Simple and standard memo
 - Reverse memo and Chronological memo
 - Linear memo and Parallel memo

(CO5 BL2)

2. Who invented Transactional model of Communication?
- Henry James
 - James Anderson
 - Dean Barunland
 - Thomas Macaulay

(CO1 BL1)

3. Verbal Communication is divided into.....
- Syllabic and Semantic Communication
 - Written and oral communication
 - Graphical and Visual Communication
 - Lingual and Kinetic Communication

(CO2 BL2)

4. In an active voice..... is an action.
- Subject
 - Object
 - Performer
 - Receiver

(CO4 BL1)

5. How many articulators are there in the English language?
- Standard and auditory articulators
 - Lingual and non-lingual articulators
 - Inverted and lateral articulators
 - Active and passive articulators

(CO3 BL1)

6. The word communication is derived from a Latin word

(CO1 BL1)

- a. Commincute
- b. Communal
- c. Communis
- d. Communicare

(CO2 BL1)

7. Vertical Communication is divided into.....
- a. Face to face and virtual communication
 - b. Upward and downward Communication
 - c. Horizontal and diagonal communication
 - d. All of these

(CO 4 BL)

8. In a portfolio, what is the primary purpose of the "Reflection" section?
- a. to list academic qualifications
 - b. to showcase professional experience
 - c. to provide a critical self-analysis of learning experiences
 - d. To display personal hobbies and interests

(CO3 BL1)

9. What are the main needs to addressed in discussions?
- a. Regulatory and expressive needs
 - b. Informational and regulatory needs
 - c. Professional and Social needs
 - d. Identity and aesthetic needs

10. In a chronological resume, how is the work experience section typically organized? (CO5 BL2)
- a. alphabetically by job title
 - b. based on the relevance to the current position
 - c. randomly, emphasizing diverse skill sets
 - d. according to the chronological order of employment, starting with the most recent

Q2. Attempt any three questions from the following.

3*5 =15 Marks

1. Compare and contrast between Verbal and non-verbal skills of Communication. (CO 1,2 BL4)
2. Describe the impact of colonial history on the development of English in India. (CO4 BL2)
3. Summarize the key skills that needs to be kept in mind during an interview. (CO3 BL2)
4. Outline the functions served by a well-crafted resume. (CO5 BL2)

Q3. Attempt any one question from the following.

5*1= 5 Marks

1. Enumerate the key aspects of ACT Communication model and its limitations along with diagram of the model. (CO1, 2, 3 BL4)
2. Assess the effectiveness of different sections in an e-Portfolio and determine their impact on presenting a candidate's qualifications (CO 4,5 BL4)
